
What Have EEG Foundation Models Actually Learned About the Conscious Brain?

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Abstract

EEG foundation models (FMs) report strong performance on standard benchmarks, but whether their embeddings carry clinical information beyond what hand-crafted markers already expose remains untested in out-of-distribution clinical populations, precisely the data-scarce, heterogeneous settings where FMs hold the most promise. Using disorders of consciousness as a case study, we evaluate five FMs (BIOT, LaBraM, EEGPT, NeuroLM, CBraMod) on 300 recordings from 249 patients along two complementary axes: zero-shot clinical utility against a validated domain-knowledge (DK) baseline across six classification tasks, and a representational audit combining layer-wise linear probing of DK marker families, fold-internal residualisation, and Mutual k-Nearest Neighbours (MKNN) alignment. Several FMs (namely NeuroLM, CBraMod, and LaBraM) match or exceed DK-level performance across clinical discrimination tasks despite being fully out-of-distribution. FM embeddings faithfully recover the spectral and information-theoretic content validated by decades of consciousness research, with LaBraM and NeuroLM additionally capturing non-trivial evoked structure. Regressing out the linearly DK-predictable component shows that the FM clinical signal substantially overlaps with established markers, while a meaningful residual margin survives in certain models and tasks, pointing to information either orthogonal to, or non-linearly related to, the DK. FMs reach MKNN alignments up to 0.32 with the DK space, meaning that the way patients are arranged relative to one another in FMs' latent spaces only partially overlaps the DK structure. Together, these results show that current FM embeddings already span much of the clinically validated DK content while opening additional representational axes: the most promising horizon for translating scale into clinical value.

1 Introduction

Disorders of consciousness (DoC) encompass a spectrum of severely altered states of awareness following acute brain injury, ranging from coma and unresponsive wakefulness syndrome (UWS) to minimally conscious state (MCS) [1]. Accurate diagnosis is clinically critical given its profound consequences for prognosis, therapeutic decisions, and end-of-life decisions. Electroencephalography (EEG) is particularly well-suited for this population given its non-invasive nature, bedside applicability, and sensitivity to the cortical dynamics that index residual consciousness. A growing body of work has established quantitative EEG biomarkers sensitive to consciousness level, spanning spectral power, connectivity, auditory-evoked potentials, and information-theoretic complexity [2, 3, 4, 5].

In parallel, the machine learning community has witnessed the rapid development of EEG foundation models (FMs): large pretrained architectures designed to transfer across domains with minimal or no task-specific fine-tuning. Models such as BIOT [6], LaBraM [7], EEGPT [8], NeuroLM [9] and CBraMod [10] have demonstrated strong performance on standard benchmarks including motor

imagery classification, sleep staging, and emotion recognition. These results have fuelled optimism about the potential of FMs to accelerate EEG analysis in clinical neuroscience.

This trajectory is, implicitly, a bet on Sutton’s “bitter lesson” [11]: that general methods leveraging computation and scale eventually outperform hand-crafted domain knowledge (DK). In the context of EEG, this means that a large enough corpus of neural recordings, paired with a sufficiently expressive model, will outperform feature-engineered markers (accumulated over decades of clinical neuroscience) on diagnostic tasks. In the regimes where this lesson has been tested: large labelled corpora and in-distribution benchmarks, the evidence is encouraging. Nevertheless, where the bitter lesson starts to fall short is in disciplines where data is scarce, noisy, and heterogeneous, and where data-driven decisions demand interpretable justification beyond predictive performance; and precisely this is where clinical neuroscience sits.

There are two gaps that separate the scientific fascination of FMs achieving strong predictive performance on brain data from their utility in clinical settings. Firstly, current EEG FM evaluations live adjacent to their pretraining distributions: TUEG [12] trained models are tested on TUAB and TUEV [13], SEED trained models are tested on SEED-like emotion tasks, motor-imagery models on motor-imagery benchmarks [14], and the pretraining corpora overlap substantially across models. Thus, there is no assurance that the high predictive performance seen in the literature will transfer to smaller, out-of-distribution (OOD) clinical populations.

Second, even when an FM matches or exceeds a hand-crafted baseline, the question of *what its embedding has learned* remains largely untouched. While there have been initial analyses to interpret these models [15, 16], these remain limited in scope. To the best of our knowledge, no prior work has systematically examined whether FM representations merely capture the same physiological information already exposed by standard clinical features. Addressing this gap is an absolute prerequisite for the accountable use of FM embeddings in any clinical setting. This matters on two fronts: diagnostic and prognostic decisions cannot rest on opaque predictions, but must be anchored in identifiable physiological content that clinicians can verify; and, beyond accountability, characterising what FMs encode opens the possibility of surfacing clinical structure that established markers do not yet capture, turning these models from prediction engines into instruments for generating new neurophysiological knowledge.

As far as we can determine, this paper presents the first evaluation framework that attempts to fill in both mentioned gaps in EEG foundation models, using as case study the DoC patient population. We evaluate five EEG-FMs (BIOT, LaBraM, EEGPT, NeuroLM, CBraMod) against a DK baseline [5] along two complementary dimensions.

1. **Utility on OOD clinical data.** Several FMs match or exceed DK-level performance zero-shot. On CRS-R Diagnostic, LaBraM and CBraMod reach DK-level performance and NeuroLM significantly exceeds it, despite never having seen a DoC patient during pretraining. Individual FMs further outperform the DK on dimensions it was not designed for: BIOT, LaBraM, and CBraMod on Delay (UWS), and NeuroLM on Etiology (MCS).
2. **Representational audit.** (i) Layer-wise linear probes confirm that FMs faithfully recover the DK’s spectral and information-theoretic axes, with LaBraM and NeuroLM further capturing evoked structure. (ii) Once the DK-predictable component is linearly removed, FM performance collapses toward chance on most tasks; the cases in which residual signal survives suggest these contain genuinely orthogonal clinical information, or non-linearly encoded DK information. (iii) No FM exceeds an MKNN alignment of 0.32 with the DK space, revealing that linear marker recoverability and FM patients’ geometric alignment with the DK latent space are related but non-equivalent properties.

Together, these results map a representational landscape where FMs already capture much of the content of established clinical markers and additionally span complementary axes.

2 Experimental Set Up

2.1 Dataset

We analyse 300 recordings from 249 DoC patients at a single clinical centre. In each recording, patients were assessed at least three times with the Coma Recovery Scale-Revised (CRS-R) [17],

which yields a score in [0,23] over a hierarchical battery of auditory, visual, motor, oromotor, communication, and arousal items. Each recording’s CRS-R diagnosis (UWS or MCS) was assigned based on the highest CRS-R score across these assessments. Beyond the diagnostic label, the dataset includes the patient’s **etiology** (restricted here to anoxia or traumatic brain injury, considering the subset of patients with such conditions) and the **delay** since injury, binarised as acute or chronic. The cohort comprises 144 UWS and 156 MCS recordings, with 40.7% anoxic and 25.7% TBI etiologies, and 67.3% acute versus 32.7% chronic recordings.

All patients underwent the local-global auditory oddball paradigm [18, 19]: a block-structured protocol in which sequences of five tones establish a local regularity, while the fifth tone either preserves or violates a (global or local) regularity across sequences. The paradigm evokes a hierarchy of stimulus-locked components associated with auditory deviance detection at multiple timescales, making it one of the most established paradigms for probing residual conscious processing in non-communicating patients.

EEG was recorded with EGI systems with 256 electrodes. Preprocessing was performed with a configuration-driven MNE-Python pipeline applying a 0.5–45 Hz bandpass, a 50 Hz notch, epoching, automatic artifact rejection (across channels and epochs), bad-channel interpolation, and common-average re-referencing. Epochs were extracted around stimulus onset, and DK markers were computed with in the same way Engemann et al. describe in their work [5].

2.2 DK baseline

Our baseline reimplements most of the markers of Sitt et al. [2], who systematically screened EEG-derived measures for their capacity to discriminate states of consciousness, and Engemann et al. [5], who consolidated this work into a multivariate classifier validated across institutions and EEG configurations.

The markers span four families: **(i) Spectral (14 markers)**: absolute and normalised band powers in δ , θ , α , β , and γ , plus median spectral frequency and spectral entropy at three thresholds (SE, SE₉₀, SE₉₅); **(ii) Information theory (5 markers)**: permutation entropy in θ , α , β , γ , and Kolmogorov complexity; **(iii) Connectivity (4 markers)**: weighted symbolic mutual information [3] in θ , α , β , γ ; **(iv) Evoked (4 markers)**: contingent negative variation and the auditory potentials P1, P3a, and P3b.

Each marker is summarised across channels and epochs following one of the four schemes in Engemann et al. [5]: an 80%-trimmed mean across trials and time, then a mean across sensors, yielding a single value. This produces a 27-dimensional embedding per recording. Crucially, every dimension is interpretable by construction, corresponding to a specific marker whose clinical meaning has been established across years of DoC research.

2.3 Foundation models

Model selection. We evaluate five pretrained EEG foundation models (chosen among the most highly cited in the literature) to span three dimensions: (1) architectural family: neural transformer (LaBraM [7]), biosignal transformer (BIOT [6]), dual self-supervised transformer (EEGPT [8]), criss-cross transformer (CBraMod [10]), and LLM-aligned encoder (NeuroLM [9]); (2) pretraining scale, from ~ 1 k to 25k hours; and (3) pretraining objective. Table 1 summarises their key properties.

Table 1: Key properties of the five evaluated models: BIOT, LaBraM-B, EEGPT, NeuroLM-B, CBraMod. Models are ordered from oldest (first row) to most recent (last row).

Model	Pretraining (hrs)	Params	Pretraining objective
BIOT	$\sim 5,000$	3.3M	Supervised joint pretraining
LaBraM-B	$\sim 2,534$	5.8M	Masked reconstruction
EEGPT	$\sim 1,200$	10M	Masked reconstruction + spatio-temporal alignment
NeuroLM-B	$\sim 25,000$	254M	Autoregressive reconstruction
CBraMod	$\sim 9,000$	4M	Masked reconstruction

Model-specific preprocessing. Each model defines its own input format: sampling rate, channel subset, window length, and patch size. We reproduce each specification exactly on our cohort, preserving as many of the original 256 channels as possible while maintaining trial structure by aligning discrete tokens or patches to single epochs (via preprocessing or padding). In particular, channel subsampling and spatial remapping were required only for BIOT and EEGPT, whose native input formats differ from our high-density EGI-256 montage. BIOT operates on 18 channels: 16 bipolar derivations between adjacent 10–20 electrodes plus two mastoid-referenced channels, whereas EEGPT operates on a 60-channel referential 10–10 layout. We projected the EGI-256 data onto these two target layouts in different ways, dictated by how each layout aligns with the EGI net. For BIOT, all 18 required sites coincide with specific sensors of the EGI HydroCel GSN-256 net. We selected these EGI sensors directly and then computed the 18 bipolar differences; no spatial interpolation or re-referencing was applied. For EEGPT, not all sites in the layout align with individual EGI sensors, so each was reconstructed by averaging the surrounding cluster of EGI electrodes.

Inner-layer probing. Activation hooks were placed at multiple intermediate layers across the five FMs, at architecturally meaningful points such as patch embeddings, intermediate transformer blocks, and final output layers. Probe points are therefore heterogeneous across models, reflecting each architecture’s structural constraints while ensuring full coverage of representational depth. Activations were mean-pooled into a single fixed-dimensional embedding (matching that layer’s embedding dimension), yielding one embedding per recording per layer.¹

2.4 Evaluation

The evaluation has two parts, each with its own questions, protocols, and set of metrics.

Part I: Utility. The first dimension asks *How do EEG FMs perform compared to the DK baseline on DoC patients?* Concretely, we evaluate whether frozen, off-the-shelf FM embeddings yield clinical classifiers competitive with the DK across six tasks: (i) UWS versus MCS diagnosis from the CRS-R; (ii) Etiology, stratified by consciousness state (UWS and MCS separately) to prevent overestimation from state confounding; and (iii) Prognosis (Improvement) at 6 months, 1 year, and 2 years, where improvement means progressing to a higher level of consciousness than at the time of recording.

For every task and FM, we use stratified nested cross-validation (CV) with 5 folds and 20 repeats, with recordings from the same patient always assigned to the same fold to prevent leakage, and identical fold splits applied across all models. The classifier choice (Support Vector Machine, Multi-Layer Perceptron, Kernel Ridge, Random Forest, or XGBoost) is treated as a hyperparameter, and a joint search is performed over both the classifier family and its associated hyperparameters: for each outer fold, the optimal configuration is selected on inner-fold performance and then evaluated on the held-out outer fold. We report the mean ROC AUC across outer folds and repeats; AUC distributions between the DK and each FM are compared via a t-test with p -values FDR-adjusted (Benjamini-Hochberg).

Part II: Representation. The second dimension shifts from performance to representation, organised around three progressively deeper questions. The first: *which markers are encoded, and how does this information emerge across layers?* We use linear probes [20]: for every layer and marker pair, we train a ridge regression to predict the marker from the layer embedding under stratified 5-fold CV (patient-grouped, as above) and report R^2 , yielding a full map of marker decodability across depth. We also evaluate each layer’s embedding as a feature space for the CRS-R diagnostic task, following the same nested CV protocol as in the Utility dimension. This produces a layer-wise AUC profile that directly bridges representational content and clinical utility.

The second question: *does the DK + FM combination add value, and is there predictive information in FM embeddings beyond what known markers explain?* We first concatenate each FM embedding with the DK feature vector and retrain under the same nested CV protocol, asking whether the combination improves over either component alone. We then perform a residualisation analysis carried out strictly fold-internally to avoid leakage [21, 22]: within each outer fold, a ridge regression of the FM embedding on the DK features is estimated on the training subjects only, and the fitted model is then used to predict and subtract the DK-explainable component from both training and

¹A detailed list of extracted layers per FM can be found in Appendix B

test embeddings. Standardisation parameters and subject-grouped splits handled identically. This out-of-sample deconfounding protocol prevents the inflated estimates that arise when the regression is fit jointly on train and test [23]. A downstream classifier is then trained on training-fold residuals and evaluated on test-fold residuals only. Corrected t-tests with Benjamini-Hochberg FDR correction, assess FM + DK against FM alone, and FM residualised against FM alone.

The third and final question: *how aligned is the geometry of the DK space with that of the FMs?* We compute the Mutual k -Nearest Neighbours alignment metric (MKNN, $k = 20$) [24] between the DK and FM representations of each recording, reporting the overlap of neighbourhood sets as a measure of global geometric correspondence at the patient level. A chance-level reference is built by permuting the rows of the DK matrix 1000 times while leaving the FM matrix intact, recomputing MKNN on each permuted pair.

Classical evaluation (linear probing, repeated nested CV, MKNN) runs CPU-only on the institutional HTCondor cluster; FM inference uses a single GPU. The full compute budget is described in Appendix A. Anonymised code is available at <https://anonymous.4open.science/r/2F7B/README.md>.

3 Results

3.1 Utility

Various EEG FMs match or exceed the DK baseline classification performance, establishing FM embeddings as viable features that complement hand-crafted clinical markers. The performance of the EEG FMs across the clinical metrics of interest is shown in Fig. 1. Each subplot corresponds to a specific metric, with the EEG FMs displayed along the x-axis and their performance, measured by AUC, on the y-axis.

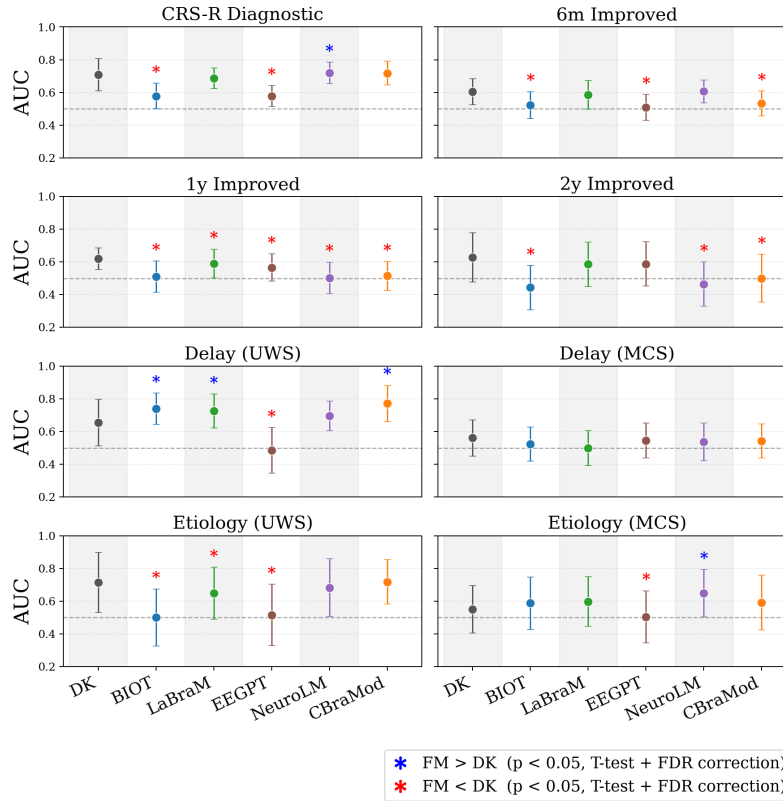


Figure 1: Nested cross-validation AUC results of EEG FMs and DK on predicting six clinical metrics.

First, on the diagnostic and prognostic tasks (top two rows), several FMs approach DK-level performance, with the DK retaining an edge in some configurations. Two results stand out as particularly encouraging: NeuroLM outperforms DK on CRS-R and matches it on 6m prognosis, plus LaBraM matches DK on CRS-R Diagnostic and on both 6m and 2y prognosis. These are non-trivial results given that the task is fully OOD. Notably, NeuroLM’s neural tokenizer is built directly on LaBraM’s VQ encoder, suggesting that the shared backbone may underlie their convergent zero-shot competence on DoC. On the other hand, BIOT shows the largest gap with DK across these four tasks. Rather than reflecting a limitation of electrode count, this gap more plausibly stems from its supervised joint pretraining objective, which may yield representations more narrowly tuned to its pretraining tasks and less transferable to fully OOD targets like DoC.

The Delay and Etiology tasks tell an equally compelling story. On Delay (UWS), three FMs (BIOT, LaBraM, and CBraMod) significantly outperform DK, and on Etiology (MCS), NeuroLM demonstrates a similarly significant edge. This pattern is striking in its own right, and its interpretation deserves emphasis: the DK markers were never designed to discriminate delay or etiology, they were specifically engineered to support the diagnostic and prognostic predictions captured in the first two rows. The fact that FMs, equally agnostic to these targets during training, match or surpass the baseline along these very dimensions points to a remarkable richness in their learned representations, one that extends well beyond what their pretraining objectives would have explicitly encouraged.

Taken together, these results paint a coherent picture of viability rather than dominance. Despite being entirely OOD and never having been exposed to a DoC patient during pretraining, frozen FM embeddings consistently match or exceed DK-level performance across a range of clinically meaningful discriminative tasks. This establishes FM representations as competitive off-the-shelf features for consciousness-related clinical prediction and, crucially, raises the question of why they succeed. The representational analysis that follows is motivated precisely by this question: what structure in these embeddings actually underlies such performance.

3.2 Representation

FM embeddings faithfully recover the spectral and information-theoretic content, and the strongest models (LaBraM, NeuroLM) further capture evoked structure. To unveil the structure of the learned representations, we trained linear ridge regression probes on the embeddings of every EEG FM at multiple intermediate layers, quantifying how much information each layer retains about the four DK marker families. Results are shown in Fig. 2: the first four rows display the cross-validated R^2 between probe-predicted and ground-truth marker values, one row per family (Evoked, Connectivity, Information Theory, Spectral), while the fifth row reports the CRS-R diagnostic AUC obtained from a linear classifier trained on the same layer-wise embeddings.

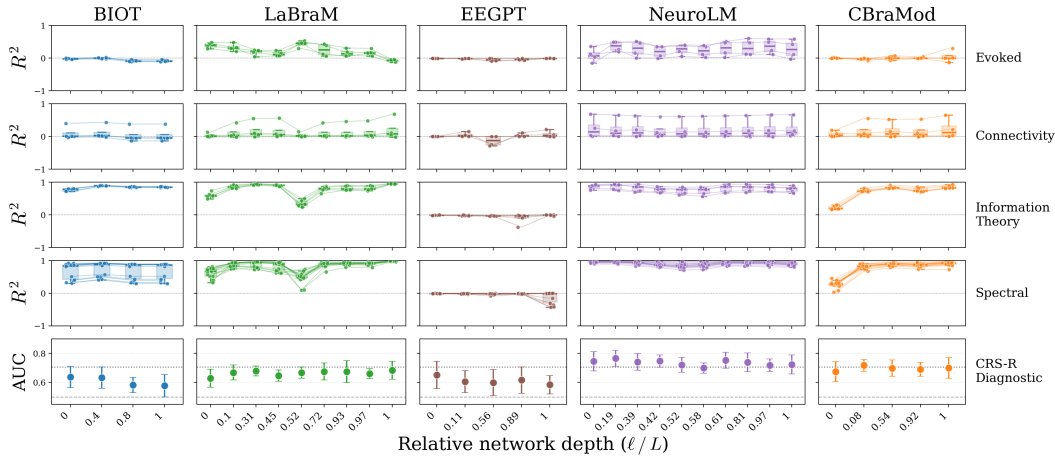


Figure 2: Linear probing and classification results across the FMs layers. The first four rows display the marker information encoded across layers for all the FMs. Markers’ families are specified at the right y-axis. Detailed results of the linear probing R^2 scores per markers and per layers can be found in Appendix B. The last row shows classification performance of the FMs’ embedding at each layer.

Three main trends emerge from Fig. 2. First, across most models (EEGPT excepted) the most linearly encoded families are Spectral and Information Theory. This is expected: both reflect global, stationary properties of the signal, naturally captured by self-supervised objectives on raw waveforms. Connectivity and Evoked markers are recovered to a lesser extent, and this too is interpretable. Connectivity requires joint modelling of cross-channel relationships that none of these models is trained to preserve, while Evoked markers are time-locked to specific stimulus events within the local-global paradigm: an experimental structure entirely absent from all five pretraining corpora. One striking exception stands out: both LaBraM and NeuroLM exhibit non-trivial Evoked R^2 , indicating that these two models have implicitly captured aspects of stimulus-locked responsiveness despite never having encountered such structure during pretraining.

Second, the encoded information is largely sustained across layers, with one exception: CBraMod shows a subtle monotonic increase in both Spectral and Information Theory R^2 with depth, suggesting that its criss-cross architecture progressively refines spectral and entropic representations through the transformer stack. Despite this enrichment, the CRS-R classification AUC remains essentially flat across all layers and all models. This is unsurprising: since none of these FMs were optimised for DoC classification, no clinical axis is available to guide such refinement, and depth alone cannot reorganise the geometry toward consciousness-related discrimination.

Third, EEGPT stands out as the weakest model across virtually all marker families, with R^2 near zero at every layer. A plausible explanation is that its spatio-temporal masked reconstruction objective, combined with a patch-level architecture, favours local temporal dynamics over the broadband spectral and complexity features the DK markers reflect.

In sum, these probing results indicate that FMs already recover much of the spectral and information-theoretic content validated by consciousness research, with LaBraM and NeuroLM extending into evoked structure that no pretraining objective directly targeted. A more striking observation is that in CBraMod classification performance remains stable across depth even at early layers, suggesting that its clinical signal is not fully reducible to these marker families. This motivates the residualisation analysis in the next section.

Across our clinical tasks, foundation models reveal three regimes—mediated by, complementary to, and partially orthogonal to the linear information carried by classical neurophysiological markers. To interrogate this relationship between FM representations and the clinical information encoded by the DK, we conducted two complementary experiments whose results are shown in Fig. 3. Each subplot corresponds to one clinical task; FDR corrected t-tests were used to assess significant differences between FM alone, FM + DK, and FM residualised.

Three distinct regimes emerge across models and tasks. The first appears in models whose zero-shot performance is already competitive with the DK, seen most clearly in NeuroLM and LaBraM on CRS-R and Etiology (MCS). Here, concatenating the DK leaves performance essentially unchanged while residualisation significantly degrades it, indicating that the FM’s clinical utility builds on the linear physiological content validated by the DK. In some cases, however, a slim discriminative capacity survives residualisation, which may reflect genuinely orthogonal clinical information the DK does not parameterise, or DK-related physiology encoded non-linearly by the FM.

The second regime is complementary and is most visible in EEGPT and BIOT on Etiology (UWS) and CRS-R Diagnostic, where FM + DK substantially outperforms FM alone, indicating that the combination injects missing physiological information, yielding genuine complementarity. Regressing out linear DK information drops performance to near chance or significantly below FM alone, confirming that the FM itself contributes little independent clinical signal.

The third regime arises when all three conditions remain comparable, with neither DK augmentation nor residualisation deviating substantially from FM alone. This typically reflects a weakness in the DK’s own discriminability for the task, as in Delay (MCS), where the DK itself performs near chance.

A methodological caveat applies to apparent gains from residualisation. In some (FM, task) pairs the residualised FM scores numerically above FM alone (e.g. CBraMod on 2y prognosis, BIOT on Delay (MCS)). Such increases should not be read as suppressor structure: as shown by Hamdan et al. [25], pairing linear residualisation with a non-linear downstream classifier can cause confound-leakage, whereby the classifier reconstructs the supposedly removed confound through interactions among

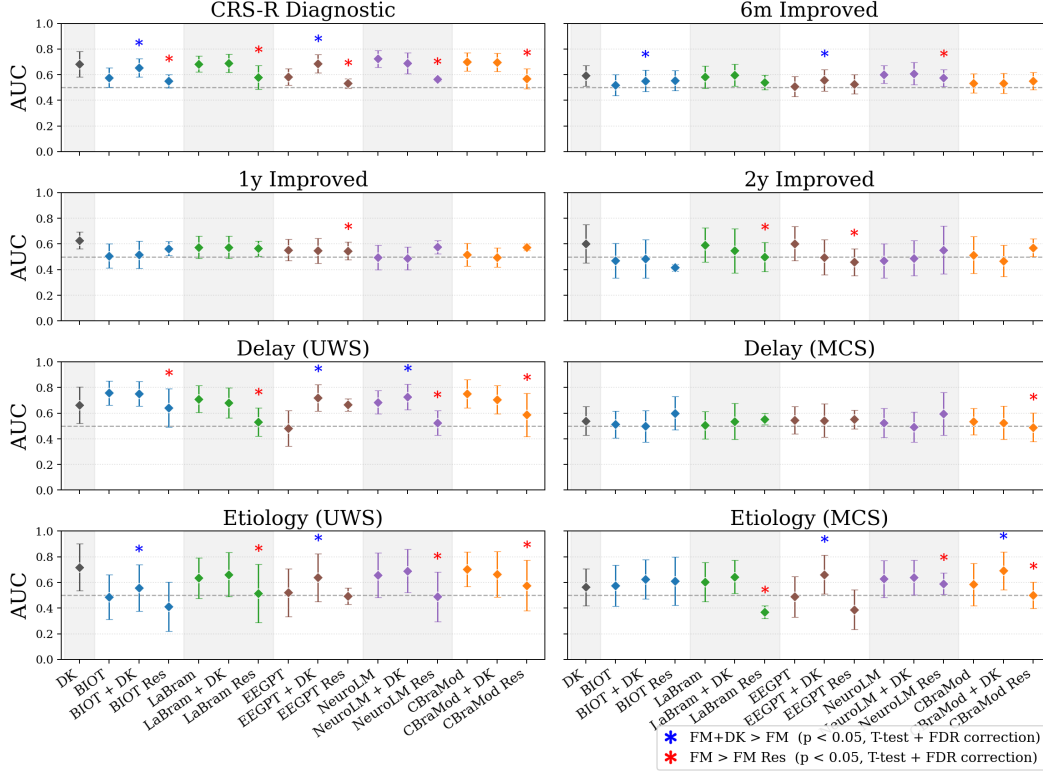


Figure 3: AUC performance across different clinical metrics for DK, FMs, FMs combined with the DK and the FMs residualized.

residualised features, inflating apparent performance. We therefore treat residualisation as a one-sided diagnostic, uninformative when it improves on FM alone.

Taken together, the three regimes trace a coherent landscape. The dominant axis of clinical information across our tasks is the one already spanned linearly by classical neurophysiological markers: strongly DK-aligned FMs (NeuroLM, CBraMod, LaBraM) derive their utility primarily from this shared content, yet a residual signal often survives residualisation, suggesting that self-supervised pretraining can also encode prognostic structure beyond a linear projection of the DK. Weakly aligned models (eg. EEGPT) only become competitive once augmented with the DK, and residualisation collapses their performance toward chance. The value of current FMs for clinical EEG therefore lies in a dual contribution: faithfully encoding validated neurophysiology, while spanning new dimensions or re-expressing known ones in forms the linear DK cannot reach. This positions FMs learned representations as natural complements to, and potential extensions of, established clinical markers. Future studies explicitly designed to disentangle these contributions will be essential to clarify where, and how, FMs add clinical value beyond established markers.

FM and DK embeddings occupy partially overlapping but structurally distinct axes: even the best-aligned FM reproduces only a fraction of the DK’s neighbourhood structure, and global alignment dissociates from marker linear recoverability. To assess whether the correspondence between FM and DK representations extends beyond individual marker encoding to the global organisation of the patient-level embedding space, we computed the MKNN alignment metric ($k = 20$) between each FM’s embedding and the DK feature space. Results are shown in Table 2, alongside a permutation-based chance baseline $\pi(\text{DK})$. An additional ablation study showing MKNN alignment results for different values of k can be found in Appendix C. These show that the relative alignment results across FMs are preserved independently of the value of k .

We can see that four out of five FMs exceed the ~ 0.07 chance baseline by substantial margins, confirming that their embedding geometries carry meaningful DK structure well above random correspondence. Two further observations deserve notice. First, even the best-aligned model reaches

only 0.319, meaning fewer than half of a patient’s nearest neighbours in DK space are also nearest neighbours in FM space. Second, the MKNN ranking only partially mirrors the probing R^2 and downstream utility results: NeuroLM, despite capturing much of the DK content linearly, sits in the lower half of the ranking, while BIOT, whose per-marker R^2 is near-zero in several families, emerges as second most aligned. This dissociation suggests that linear marker recoverability and global geometric alignment are related but non-equivalent properties: a model can encode the *content* of the DK without reproducing its *organisational structure*, and vice versa, opening a productive question about which axes of patient variance current markers do not yet parameterise.

Taken together, these results confirm that FMs reproduce a substantial portion of the DK geometry while organising patient variance along complementary axes. This structural divergence stresses the need to characterise what clinically interpretable structure FM embeddings encode beyond what is linearly recoverable from established markers, potentially revealing latent representations absent from current DK—dimensions that could sharpen our understanding of pathophysiology and clinical decision making.

Table 2: MKNN alignment metric results for each FMs with the respective baseline defined by $\pi(DK)$.

	BIOT	LaBraM	EEGPT	NeuroLM	CBraMod
MKNN	0.201	0.319	0.087	0.149	0.186
$\pi(DK)$	0.068	0.068	0.073	0.069	0.068

4 Conclusion

On a fully OOD cohort of disorders-of-consciousness patients, EEG FMs demonstrate clear viability: leading models match or outperform the DK baseline on zero-shot classification tasks, even though no model had ever been exposed to a DoC patient during pretraining. Our representational audit organises this performance into three regimes—mediated, complementary, and partially orthogonal—that clarify when FMs lean on classical neurophysiology and when they reach beyond it. Much of the clinical signal in the strongest FMs traces back to content the DK already encodes explicitly, and once that overlap is removed, performance falls across most tasks. What survives is slim but meaningful: a margin of discriminability that classical markers do not reach linearly, and the most promising horizon for the clinical contribution of these models. FMs further reorganise patient variance along axes the DK does not span, showing that linear marker recoverability and geometric alignment are related but non-equivalent properties. Translating these contributions into interpretable clinical signal will reveal whether scale can extend, rather than merely echo, decades of neuroscience. As pretraining corpora diversify and FMs continue to scale, we expect them to surface patterns current markers have yet to capture.²

5 Limitations and Future Work

Four limitations bound these results. First, channel configurations are unmatched: DK markers use the full 256-channel EGI montage while BIOT and EEGPT operate on 18 and 60 electrodes, so part of their gap may reflect sensor count rather than representational quality. Second, the analysis draws on a single clinical site; despite a cohort large by DoC standards, the absence of external validation limits generality. Third, the representational audit relies on linear probes and residualization, leaving non-linear and geometric structure uncharacterized. Finally, FMs are evaluated frozen, zero-shot, and mean-pooled, with no robustness checks against clinically realistic perturbations (channel dropout, electrode noise, motion artifacts). These motivate four matching directions: channel-matched DK controls to disentangle representational quality from sensor count; external validation on independent DoC cohorts; non-linear probing and deconfounding with formal geometric characterization of FM-specific axes; and perturbation stress-tests alongside fine-tuning, parameter-efficient adaptation, and alternative pooling. Together, these would consolidate the present framework into a reusable methodology for auditing EEG foundation models before clinical deployment.

²This mirrors a trend seen in ML a decade ago, when CNNs replicated and ultimately outperformed handcrafted kernels for computer vision tasks.

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A Compute

FM feature extraction runs on a single GPU and takes $\approx 1\text{--}6$ h per model across the full cohort of 300 recordings, depending on architecture size and patch resolution. For each FM we cache both pooled embeddings and between 4 and 9 transformer-layer embeddings. None of the downstream evaluations require a GPU. FM-embedding classification (repeated nested CV with 5 outer folds and 20 repeats, per-fold selection over 5 classifiers, six clinical tasks) runs on an 8-core node in $\approx 10\text{--}16$ h per FM; the same configuration is rerun for the DK+FM combined and FM-residualised variants. Layer-wise linear probing (R^2 ridge regression for every (layer, marker) pair and nested-CV CRS-R classification at each stored layer) runs in $\approx 12\text{--}16$ h per FM on an 8-core node. The MKNN alignment with its 1000-permutation null finishes in $\approx 1\text{--}2$ h on a single 8-core node. The project total estimated from cluster accounting (Table 3) is $\approx 6,000$ effective CPU-hours and ≈ 35 single-GPU (P100) hours over ≈ 130 jobs, including preliminary and failed experiments not reported in the paper.

Table 3: Total project compute estimated from HTCondor accounting, broken down by experiment family. Includes preliminary and failed runs; the experiments reported in the paper are a subset of these totals. CPU-hours are wall-clock $\times$ allocated cores on 8–16-core jobs with 16–48 GB RAM; GPU-hours are wall-clock single-GPU time on NVIDIA Tesla P100 (16 GB) nodes.

Experiment family	CPU-hours	P100-hours	# jobs
FM feature extraction	≈ 40	≈ 22	5
FM-embedding classification	≈ 540	0	30
Layer-wise linear probing (classification and regression)	≈ 580	0	5
DK-marker baseline and DK+FM combined	≈ 600	0	36
FM-residualised classification	≈ 540	0	30
MKNN alignment (with permutation null)	≈ 15	0	1
Preliminary, ablation & failed runs	$\approx 3,700$	≈ 12	≈ 25
Total	$\approx 6,000$	≈ 35	≈ 130

B Linear Probing Results

In the following tables we see the linear probing R^2 scores for each marker across all FMs. The column’s names indicate the layers where the probing was done on each model, with the specific relative depth they are in.

Table 4: R^2 per marker $\times$ layer for BIOT.

Marker	pre_transformer (0)	transformer_1 (0.4)	transformer_3 (0.8)	Last (1)
ContingentNegativeVariation/default	-0.040	-0.023	-0.115	-0.115
KolmogorovComplexity/default	0.788	0.899	0.840	0.840
PermutationEntropy/alpha	0.794	0.902	0.864	0.864
PermutationEntropy/beta	0.711	0.871	0.835	0.834
PermutationEntropy/gamma	0.721	0.883	0.850	0.850
PermutationEntropy/theta	0.805	0.900	0.867	0.867
PowerSpectralDensity/alpha	0.307	0.426	0.326	0.326
PowerSpectralDensity/alphan	0.913	0.916	0.869	0.869
PowerSpectralDensity/beta	0.327	0.405	0.297	0.297
PowerSpectralDensity/betan	0.880	0.915	0.869	0.869
PowerSpectralDensity/delta	0.385	0.494	0.391	0.391
PowerSpectralDensity/deltan	0.865	0.908	0.876	0.876
PowerSpectralDensity/gamma	0.511	0.574	0.517	0.517
PowerSpectralDensity/gamman	0.849	0.869	0.875	0.875
PowerSpectralDensity/summary_se	0.854	0.915	0.883	0.883
PowerSpectralDensity/theta	0.401	0.511	0.426	0.426
PowerSpectralDensity/thetan	0.913	0.921	0.881	0.881
PowerSpectralDensitySummary/summary_msf	0.805	0.850	0.816	0.816
PowerSpectralDensitySummary/summary_sef90	0.855	0.913	0.891	0.891
PowerSpectralDensitySummary/summary_sef95	0.828	0.912	0.888	0.888
SymbolicMutualInformation/weighted_alpha	0.013	0.018	-0.045	-0.045
SymbolicMutualInformation/weighted_beta	-0.029	-0.019	-0.138	-0.138
SymbolicMutualInformation/weighted_gamma	0.009	0.033	-0.047	-0.047
SymbolicMutualInformation/weighted_theta	0.403	0.429	0.379	0.379
TimeLockedTopography/p1	-0.009	-0.020	-0.042	-0.042
TimeLockedTopography/p3a	-0.041	0.014	-0.076	-0.076
TimeLockedTopography/p3b	0.000	0.020	-0.105	-0.105

Table 5: R^2 per marker $\times$ layer for CBraMod.

Marker	patch_emb (0)	layer_0 (0.08)	layer_6 (0.54)	layer_11 (0.92)	Last (1)
ContingentNegativeVariation/default	-0.001	-0.001	-0.076	-0.006	0.006
KolmogorovComplexity/default	0.308	0.811	0.904	0.858	0.928
PermutationEntropy/alpha	0.251	0.757	0.839	0.811	0.887
PermutationEntropy/beta	0.189	0.723	0.833	0.716	0.824
PermutationEntropy/gamma	0.161	0.703	0.805	0.702	0.811
PermutationEntropy/theta	0.182	0.719	0.812	0.733	0.824
PowerSpectralDensity/alpha	0.271	0.887	0.919	0.896	0.932
PowerSpectralDensity/alphan	0.031	0.668	0.783	0.732	0.723
PowerSpectralDensity/beta	0.453	0.877	0.895	0.893	0.963
PowerSpectralDensity/betan	0.296	0.817	0.881	0.872	0.894
PowerSpectralDensity/delta	0.226	0.905	0.954	0.947	0.977
PowerSpectralDensity/deltan	0.280	0.827	0.908	0.880	0.945
PowerSpectralDensity/gamma	0.399	0.838	0.879	0.818	0.930
PowerSpectralDensity/gamman	0.263	0.801	0.856	0.836	0.888
PowerSpectralDensity/summary_se	0.316	0.832	0.912	0.891	0.946
PowerSpectralDensity/theta	0.220	0.894	0.951	0.939	0.960
PowerSpectralDensity/thetan	0.075	0.727	0.827	0.796	0.846
PowerSpectralDensitySummary/summary_msf	0.314	0.746	0.846	0.825	0.868
PowerSpectralDensitySummary/summary_sef90	0.302	0.831	0.906	0.876	0.926
PowerSpectralDensitySummary/summary_sef95	0.282	0.821	0.912	0.864	0.924
SymbolicMutualInformation/weighted_alpha	0.084	0.095	0.151	0.101	0.216
SymbolicMutualInformation/weighted_beta	-0.001	-0.002	-0.010	-0.007	-0.004
SymbolicMutualInformation/weighted_gamma	0.003	0.024	0.010	0.016	0.025
SymbolicMutualInformation/weighted_theta	0.192	0.554	0.522	0.527	0.657
TimeLockedTopography/p1	0.021	-0.082	-0.008	-0.009	-0.128
TimeLockedTopography/p3a	-0.011	-0.027	0.074	0.037	0.295
TimeLockedTopography/p3b	-0.012	-0.027	0.012	0.001	0.005

Table 6: R^2 per marker $\times$ layer for EEGPT.

Marker	patch_emb (0)	layer_0 (0.11)	layer_4 (0.56)	layer_7 (0.89)	Last (1)
ContingentNegativeVariation/default	-0.005	-0.005	-0.011	-0.017	-0.015
KolmogorovComplexity/default	-0.017	-0.017	-0.018	-0.013	-0.003
PermutationEntropy/alpha	-0.013	-0.015	-0.022	-0.019	0.002
PermutationEntropy/beta	-0.017	-0.028	-0.044	-0.031	-0.040
PermutationEntropy/gamma	-0.018	-0.031	-0.045	-0.093	0.001
PermutationEntropy/theta	-0.006	0.000	-0.037	-0.380	-0.010
PowerSpectralDensity/alpha	-0.013	-0.021	-0.015	-0.020	-0.361
PowerSpectralDensity/alphan	-0.004	-0.004	-0.011	-0.003	-0.023
PowerSpectralDensity/beta	-0.001	-0.022	-0.008	-0.018	-0.421
PowerSpectralDensity/betan	-0.015	-0.015	-0.015	-0.015	-0.006
PowerSpectralDensity/delta	-0.020	-0.022	-0.050	-0.013	-0.427
PowerSpectralDensity/deltan	-0.009	-0.003	-0.005	-0.019	-0.008
PowerSpectralDensity/gamma	-0.001	-0.036	-0.088	-0.033	-0.160
PowerSpectralDensity/gamman	-0.010	-0.010	-0.021	-0.017	-0.009
PowerSpectralDensity/summary_se	-0.013	-0.013	-0.014	-0.009	-0.006
PowerSpectralDensity/theta	-0.018	-0.018	-0.013	-0.017	-0.308
PowerSpectralDensity/thetan	-0.008	-0.010	-0.024	-0.028	-0.006
PowerSpectralDensitySummary/summary_msf	-0.010	-0.010	-0.009	-0.009	-0.008
PowerSpectralDensitySummary/summary_sef90	-0.014	-0.014	-0.027	-0.020	-0.005
PowerSpectralDensitySummary/summary_sef95	-0.015	-0.015	-0.031	-0.027	-0.002
SymbolicMutualInformation/weighted_alpha	-0.004	0.044	-0.241	0.034	0.052
SymbolicMutualInformation/weighted_beta	-0.004	-0.003	0.000	0.002	-0.001
SymbolicMutualInformation/weighted_gamma	-0.005	-0.004	-0.002	-0.001	-0.008
SymbolicMutualInformation/weighted_theta	-0.004	0.153	-0.285	0.112	0.213
TimeLockedTopography/p1	-0.013	-0.013	-0.088	-0.053	-0.004
TimeLockedTopography/p3a	-0.004	-0.027	-0.085	-0.071	-0.012
TimeLockedTopography/p3b	-0.006	-0.006	-0.006	-0.006	-0.006

Table 7: R^2 per marker $\times$ layer for LaBraM.

Marker	vqmp_enc_patch_emb(0)	vqmp_enc_block_2(0.1)	vqmp_enc_block_8(0.31)	vqmp_encode_task(0.45)	labram_patch_emb(0.52)	labram_block_5(0.72)	labram_block_11(0.93)	labram_norm(0.97)	Last(1)
ContingentNegativeVariation/default	0.482	0.482	0.241	0.235	0.527	0.390	0.182	0.162	-0.098
KolmogorovComplexity/default	0.717	0.914	0.930	0.925	0.497	0.775	0.894	0.894	0.962
PermutationEntropy/alpha	0.624	0.883	0.941	0.920	0.425	0.840	0.879	0.872	0.950
PermutationEntropy/beta	0.557	0.853	0.917	0.884	0.321	0.810	0.794	0.793	0.953
PermutationEntropy/gamma	0.491	0.808	0.904	0.866	0.285	0.769	0.752	0.738	0.940
PermutationEntropy/theta	0.579	0.812	0.941	0.906	0.231	0.655	0.794	0.766	0.952
PowerSpectralDensity/alpha	0.397	0.755	0.719	0.618	0.481	0.857	0.900	0.896	0.980
PowerSpectralDensity/alphan	0.520	0.894	0.941	0.892	0.088	0.656	0.790	0.786	0.961
PowerSpectralDensity/beta	0.581	0.746	0.745	0.608	0.738	0.888	0.928	0.926	0.975
PowerSpectralDensity/betan	0.663	0.925	0.959	0.946	0.473	0.914	0.914	0.904	0.975
PowerSpectralDensity/delta	0.519	0.826	0.781	0.669	0.618	0.960	0.944	0.951	0.984
PowerSpectralDensity/deltan	0.769	0.927	0.967	0.951	0.414	0.846	0.925	0.922	0.975
PowerSpectralDensity/gamma	0.673	0.831	0.819	0.742	0.686	0.894	0.885	0.884	0.979
PowerSpectralDensity/gamman	0.816	0.912	0.943	0.909	0.532	0.926	0.894	0.901	0.973
PowerSpectralDensity/summary_se	0.762	0.933	0.969	0.957	0.500	0.894	0.930	0.926	0.982
PowerSpectralDensity/theta	0.399	0.816	0.786	0.676	0.546	0.956	0.952	0.952	0.983
PowerSpectralDensity/thetan	0.329	0.925	0.964	0.950	0.099	0.811	0.888	0.887	0.982
PowerSpectralDensitySummary/summary_msf	0.748	0.847	0.930	0.888	0.572	0.894	0.868	0.859	0.966
PowerSpectralDensitySummary/summary_sef90	0.757	0.934	0.957	0.938	0.512	0.921	0.928	0.924	0.982
PowerSpectralDensitySummary/summary_sef95	0.713	0.935	0.958	0.937	0.472	0.904	0.915	0.913	0.984
SymbolicMutualInformation/weighted_alpha	0.019	0.049	0.105	0.068	0.024	0.032	0.026	0.044	0.114
SymbolicMutualInformation/weighted_beta	-0.015	-0.024	-0.007	-0.005	-0.011	-0.011	-0.007	-0.009	-0.016
SymbolicMutualInformation/weighted_gamma	0.004	0.016	0.050	0.045	0.007	0.017	0.018	0.030	0.048
SymbolicMutualInformation/weighted_theta	0.130	0.418	0.555	0.557	0.143	0.416	0.463	0.481	0.683
TimeLockedTopography/p1	0.431	0.326	0.044	0.069	0.462	0.059	0.084	0.064	-0.126
TimeLockedTopography/p3a	0.159	0.297	0.233	0.152	0.453	0.238	0.309	0.295	-0.050
TimeLockedTopography/p3b	0.288	0.202	0.187	0.068	0.295	0.128	0.061	0.152	-0.043

Table 8: R^2 per marker $\times$ layer for NeuroLM.

Marker	vq_enc_patch_emb(0)	vq_enc_block_5(0.19)	vq_enc_block_11(0.39)	vq_encode_task(0.42)	vq_dec_freq_block_1(0.52)	vq_dec_freq_block_3(0.58)	gpt_0(0.61)	gpt_6(0.81)	gpt_11(0.97)	Last(1)
ContingentNegativeVariation/default	0.034	0.488	0.508	0.346	0.375	0.344	0.508	0.601	0.587	0.585
KolmogorovComplexity/default	0.909	0.915	0.976	0.918	0.796	0.798	0.864	0.859	0.846	0.832
PermutationEntropy/alpha	0.890	0.913	0.857	0.784	0.775	0.775	0.862	0.853	0.847	0.810
PermutationEntropy/beta	0.819	0.787	0.732	0.707	0.667	0.651	0.714	0.738	0.712	0.709
PermutationEntropy/gamma	0.754	0.747	0.693	0.671	0.613	0.564	0.684	0.702	0.667	0.667
PermutationEntropy/theta	0.927	0.942	0.936	0.865	0.846	0.848	0.934	0.934	0.925	0.900
PowerSpectralDensity/alpha	0.963	0.970	0.959	0.932	0.911	0.918	0.958	0.956	0.957	0.939
PowerSpectralDensity/alphan	0.952	0.958	0.940	0.867	0.841	0.890	0.940	0.941	0.944	0.909
PowerSpectralDensity/beta	0.952	0.950	0.918	0.848	0.802	0.819	0.916	0.914	0.896	0.873
PowerSpectralDensity/betan	0.956	0.952	0.932	0.845	0.833	0.858	0.930	0.923	0.923	0.881
PowerSpectralDensity/delta	0.992	0.991	0.988	0.976	0.978	0.988	0.988	0.987	0.983	0.983
PowerSpectralDensity/deltan	0.980	0.979	0.970	0.940	0.922	0.922	0.975	0.972	0.966	0.955
PowerSpectralDensity/gamma	0.894	0.886	0.826	0.783	0.703	0.725	0.828	0.824	0.802	0.799
PowerSpectralDensity/gamman	0.950	0.953	0.922	0.853	0.788	0.828	0.920	0.916	0.907	0.866
PowerSpectralDensity/summary_se	0.957	0.953	0.933	0.869	0.850	0.858	0.929	0.931	0.925	0.895
PowerSpectralDensity/theta	0.985	0.986	0.977	0.968	0.965	0.967	0.978	0.980	0.981	0.970
PowerSpectralDensity/thetan	0.970	0.968	0.954	0.913	0.880	0.908	0.951	0.957	0.953	0.930
PowerSpectralDensitySummary/summary_msf	0.956	0.957	0.943	0.884	0.855	0.868	0.944	0.936	0.934	0.899
PowerSpectralDensitySummary/summary_sef90	0.939	0.938	0.902	0.841	0.804	0.815	0.888	0.895	0.890	0.854
PowerSpectralDensitySummary/summary_sef95	0.917	0.919	0.870	0.809	0.774	0.781	0.853	0.859	0.854	0.821
SymbolicMutualInformation/weighted_alpha	0.253	0.178	0.204	0.117	0.142	0.135	0.157	0.170	0.184	0.176
SymbolicMutualInformation/weighted_beta	0.001	-0.001	-0.005	-0.001	-0.002	-0.003	-0.001	-0.008	-0.009	0.020
SymbolicMutualInformation/weighted_gamma	0.033	0.033	0.029	0.026	0.021	0.018	0.031	0.029	0.030	0.058
SymbolicMutualInformation/weighted_theta	0.680	0.657	0.631	0.603	0.616	0.619	0.634	0.642	0.660	0.668
TimeLockedTopography/p1	-0.154	0.267	0.005	-0.007	0.052	0.022	0.016	0.114	0.113	-0.029
TimeLockedTopography/p3a	0.363	0.486	0.465	0.352	0.403	0.389	0.471	0.469	0.496	0.422
TimeLockedTopography/p3b	0.118	0.192	0.158	0.068	0.201	0.110	0.168	0.123	0.233	0.115

429 **C MKNN alignment across FMs for different k values**

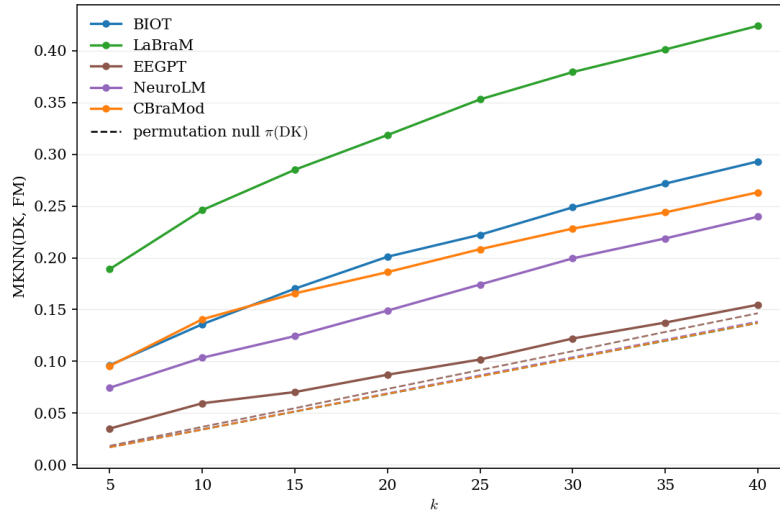


Figure 4: We see that as k increases MKNN increases consistently across models and respective baselines. Relative order is preserved for different k values, with the exception of CBraMod and BIOT for $k = 10$. Nevertheless the trend is quite continuous and smooth, allowing us to pick $k = 20$ for our analysis without further problem.

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Question: Do the main claims made in the abstract and introduction accurately reflect the paper's contributions and scope?

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Question: Does the paper discuss the limitations of the work performed by the authors?

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Justification: Section 5 explicitly discusses four limitations of our study: the channel-count mismatch between the DK baseline and the channel-constrained FMs (BIOT, EEGPT), the use of a single clinical site without external cohort validation, the reliance on linear probes and linear residualisation, the restriction to frozen zero-shot embeddings with mean pooling. We further reflect on how each of these factors may have shaped the reported results and outline corresponding mitigations in the future work paragraph.

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Justification: Section 2 fully specifies the methodology required to reproduce our results: the dataset composition and clinical labels, the preprocessing pipeline, the DK marker families and their aggregation scheme, the per-FM input requirements, the layer-wise probe extraction protocol, the evaluation tasks, the stratified subject-grouped repeated nested cross-validation, the inner-fold classifier selection, the ridge-regression linear probes, the fold-internal residualisation procedure, and the MKNN alignment with its permutation null, and the statistical tests. Compute requirements are itemised in Appendix A and an anonymised version of the evaluation repository, including environment specifications, configuration files, and instructions for running every experiment reported in the paper, is released alongside this submission at <https://anonymous.4open.science/r/2F7B/README.md> to support exact reproduction. The evaluated foundation models are publicly available under the licenses listed in question 12, and the clinical dataset cannot be redistributed for privacy reasons; however, our pipeline is fully runnable on any DoC EEG cohort with the same protocol the corresponding clinical labels.

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5. Open access to data and code

Question: Does the paper provide open access to the data and code, with sufficient instructions to faithfully reproduce the main experimental results, as described in supplemental material?

Answer: [Yes]

Justification: The anonymized code implementing our evaluation framework, including layer-wise linear probing, repeated nested cross-validation, residualisation, and MKNN alignment, is released in an anonymised public repository <https://anonymous.4open.science/r/2F7B/README.md>, together with environment specifications, the exact configuration files used for each experiment, and instructions for reproducing every result reported in the paper. The five evaluated foundation models are publicly available under the licenses listed in question 12, with the corresponding repositories cited in the main text. The clinical EEG dataset is institutional and cannot be redistributed for privacy reasons; reproducing the exact numerical results therefore requires access to the original cohort, but our code is fully runnable on any sufficiently labelled DoC EEG dataset.

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Answer: [Yes]

Justification: Section 2 specifies the dataset composition, preprocessing pipeline, the DK marker families and their aggregation scheme, and the per-FM input requirements. Compute requirements per experiment family are itemised in Appendix A, and code anonymized repository <https://anonymous.4open.science/r/2F7B/README.md>.

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Question: Does the paper report error bars suitably and correctly defined or other appropriate information about the statistical significance of the experiments?

Answer: [Yes]

Justification: All point estimates reported in Figures 1 and 3 and in Table 2 are accompanied by error bars or null distributions: AUC values are reported as mean and standard deviation across the 100 outer-fold evaluations of the repeated nested cross-validation (5 outer folds, 20 repeats), and MKNN alignment is reported with a permutation-based empirical null obtained by 1000 random row-shufflings of the DK matrix. Statistical comparisons between FMs and the DK baseline in the utility analysis, as well as pairwise comparisons between FM, FM + DK, and FM-residualised conditions, are assessed via the corrected paired t -test. The protocols for both error-bar computation and significance testing are described in Section 2, and significant differences are explicitly marked in the corresponding figures.

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Answer: [Yes]

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Justification: The authors have reviewed the NeurIPS Code of Ethics and the research conducted for this paper conforms with it. The clinical data analysed in this study were collected under the existing ethics framework of the participating institution, with appropriate institutional ethics approval and surrogate consent obtained from the legal representatives of patients unable to provide consent themselves; further details are provided in our response to question 15. EEG recordings were anonymised prior to analysis, and storage and processing complied with the relevant regional regulations governing medical data. Given the sensitive nature of these clinical recordings, the dataset will not be fully released. We use only publicly released foundation-model checkpoints under the licenses provided by their authors (see question 12), and we do not release any new model artefact that would carry dual-use, surveillance, or misuse risk.

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Answer: [Yes]

Justification: On the positive side, this work contributes a methodology for auditing EEG foundation models before clinical adoption in disorders of consciousness, a population in which diagnostic errors carry severe consequences for prognosis, therapeutic decisions, and end-of-life decisions: a more rigorous evaluation of FM embeddings against established clinical markers can help prevent the deployment of representations whose apparent performance does not reflect genuinely independent clinical signal. On the negative side, the use of FMs as black-box predictors in clinical settings carries the risk of grounding diagnostic and prognostic decisions in opaque representations that clinicians cannot verify against identifiable physiological content.

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Justification: This paper does not release any new pretrained model, generative system, or scraped dataset, so the safeguards considered under this question do not apply. We evaluate publicly available foundation-model checkpoints released by their respective authors under the licenses listed in question 12, and the clinical EEG dataset is institutional and not redistributed for privacy reasons (see question 9).

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wjq-learning/CBraMod). For software dependencies, we use MNE-Python [26] <https://mne.tools/stable/index.html> (BSD 3-clause). The clinical EEG dataset is institutional and was collected under the existing ethics framework of the participating hospital; we do not redistribute it.

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Justification: The only new asset associated with this paper is the analysis code (<https://anonymous.4open.science/r/2F7B/README.md>) implementing the evaluation framework (zero-shot classification on FM embeddings, layer-wise linear probing, fold-internal residualisation, and MKNN alignment). This code is attached in an anonymised repository together with documentation covering installation, environment requirements, the exact configuration used for each experiment, and the licence under which it is distributed (MIT);. We do not release the clinical dataset (see question 9), and we do not release new pretrained models.

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Justification: This question concerns crowdsourcing studies and active-task human-subjects research (paid annotators, human evaluators following instructions, etc.), neither of which applies to our work. Although our analysis is performed on EEG recordings from human patients, the recordings were collected as part of standard clinical assessment under the existing institutional ethics framework, with surrogate consent obtained from the legal representatives of patients unable to consent themselves.

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Justification: This research was approved by the ethics committee of the participating hospital under the protocol code routine care research, and all investigations were carried out in accordance with the Declaration of Helsinki on ethical principles for medical research involving human subjects. Because patients with disorders of consciousness are by definition unable to provide informed consent themselves, the protocol relies on surrogate consent obtained from the legal representatives of the patients in line with national regulations governing routine clinical research. The EEG recordings analysed in this study are part of standard clinical assessment, posing no additional risk beyond that of routine care; the specific institution and full protocol identifiers will be disclosed in the camera-ready version to preserve anonymity at submission.

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